

Comparison of AIM-7 Sparrow variants

Sparrow	Length (mm)	Wingspan (mm)	Launch weight (kg)	Speed (mach)	Range (km)	Warhead (kg)
AIM-7A	3740	940	65	2.5	10	20
AIM-7C	3660	1020	78	4	11	30
AIM-7D	3660	1020	78	4	11	30
AIM-7E	3660	1020	89.5	4	30	30
AIM-7F	3660	1020	105	4	70	39
AIM-7H	3660	1020	89.5	4	30	30
AIM-7M	3660	1020	105	4	30	40
AIM-7P	3660	1020	105	4	30	40

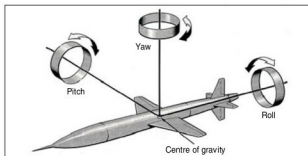


Fig. 4: Missile stabilisation



Fig. 5: AIM-7 Sparrow being launched from F-18A Hornet

The attitude control system controls the attitude of the missile in pitch, yaw and roll parameters on the designated flight path (Fig. 4). The flight path control system guides the missile to its designated target. This is achieved by determining flight path errors, generating necessary commands to correct these errors and sending these commands to the missile's control sub-system that works on servo principle.

Control units make corrective

adjustments to missile control surfaces on receiving an error signal. These units also adjust wings or fins to stabilise the missile in roll, pitch and yaw. Guidance and stabilisation are two separate processes that occur concurrently.

Radar-guided missile systems

Now, we will discuss some prominent surface-to-air and air-to-air radar-guided missile systems.

Missile systems described here include AIM-7 Sparrow, MIM-104 Patriot missile system (both semi-active radar-guided missile systems), AIM-54 Phoenix, RIM-174 ERAM,

AIM 120D AMRAAM, MBDA Exocet (all active radar-guided missile systems) and Akash surface-to-air missile.

AIM-7 Sparrow is a medium-range, supersonic-speed semi-active radar-guided air-to-air missile manufactured by Raytheon.

A derivative of this missile is the ship based version called RIM-7 Sea Sparrow used for air defence role. The missile had been in active service from 1950's till 1990's. It is be-

ing gradually phased out in favour of AIM-120 AMRAAM, an all-weather, day/night operation beyond-visual-range air-to-air missile.

AIM-7 Sparrow has a number of variants designated as AIM-7A, AIM-7B, AIM-7C, AIM-7D, AIM-7E, AIM-7F, AIM-7G, AIM-7H, AIM-7M, AIM-7N, AIM-7P and AIM-7R.

Development of AIM-7B, AIM-7N and AIM-7R was discontinued after initial interest. Table I shows a comparison of different variants of AIM-7 Sparrow. Many AIM-7 variants had ship based versions. These included RIM-7E, RIM-7F, RIM-7H, RIM-7M, RIM-7P and RIM-7R. AIM/RIM-7R programme was abandoned in 1996 due to its high costs.

The first operational use of AIM-7 Sparrow was during Vietnam war in June 1965 when the US Navy shot down two North Vietnamese MiG-17s. Due to the absence of an IFF system on the launch aircraft, long-range capability of these missiles could not be used.

First missiles had a kill probability of only 10 per cent. This problem was subsequently overcome in AIM-7E. The last use of AIM-7 Sparrow was in Operation Desert Storm in 1991, where it was extensively and successfully used on F-15 and F-16 fighter aircrafts. The missile has been on the inventory of armed forces of a large number of countries including Australia, Canada, Israel, Italy, Japan, South Korea, Spain, Taiwan, Turkey, the UK and the USA. Fig. 5 shows AIM-7 Sparrow being launched from F-18A fighter aircraft.

MIM-104 Patriot, manufactured by Raytheon, is a long-range, all-altitude, all-weather air defence system to counter tactical ballistic missiles, cruise missiles and advanced aircraft. The missile system derives its name from the radar component of the missile system.

Patriot stands for phased array tracking radar to intercept on target. It is now given the name anti-ballistic missile (ABM) system, which is its primary mission.